After Mid

Memory Interfacing

Chip Identification:

→ 2 memory element : EPROM , RAM

1. EPROM : IC 27XXX , IC 27XX
2. RAM : IC 61/62 XXX

EX-1:

IC $\boxed{27}\boxed{512}$

1st two digit memory Type — Size of memory in Kbits.

→ Memory size = 512 Kbits = $\frac{512}{8}$ KB = 64 KB

→ That's memory is EPROM 64 KB memory.

EX-2

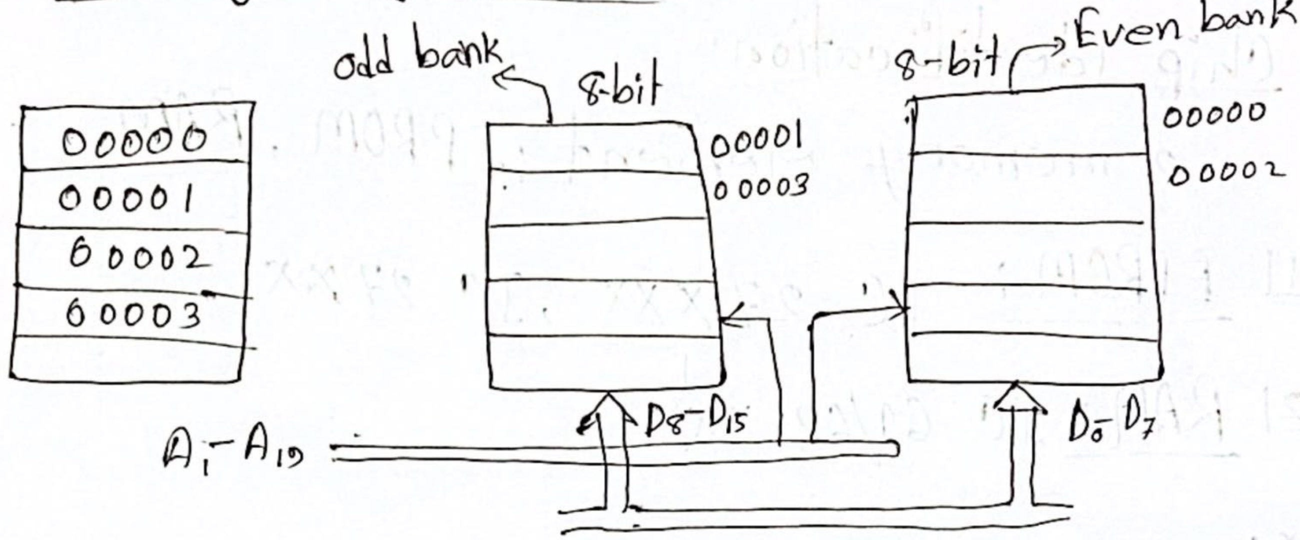
IC $\boxed{62}\boxed{64}$

1st two digit is memory type — Size of memory in Kbits.

→ Memory size = 64 Kbits = $\frac{64}{8}$ KB = 8 KB

→ That's memory is RAM 8 KB memory.

Memory Organization:



=> Absolate/ Full decoding chip selector এর সবগুলো bit নিয়ে কাজ করতে হবে।

=> Partial ~~de~~/ leaner decoding chip selector এর 2/3 টা bit নিয়ে কাজ করতে হবে।

=> Memory ৩ টি থাকলে অথবা বিজোড় সংখ্যক memory থাকলে 2 জোড়া করে কাজ করতে হবে অর্থাৎ pair করে করতে হবে।

=> If Memory Size same না এবং Memory same হলেও। অথবা Memory size same কিন্তু memory same না হলে pair করতে পারবো না। ~~এবার~~ same memory and same size হলেই 2টি memory pair করে কাজ করতে পারবো।

P-1: Interface two 4K X8 EPROM and 4K x8 RAM chip with 8086.

Soln: Given that,

Memory size of

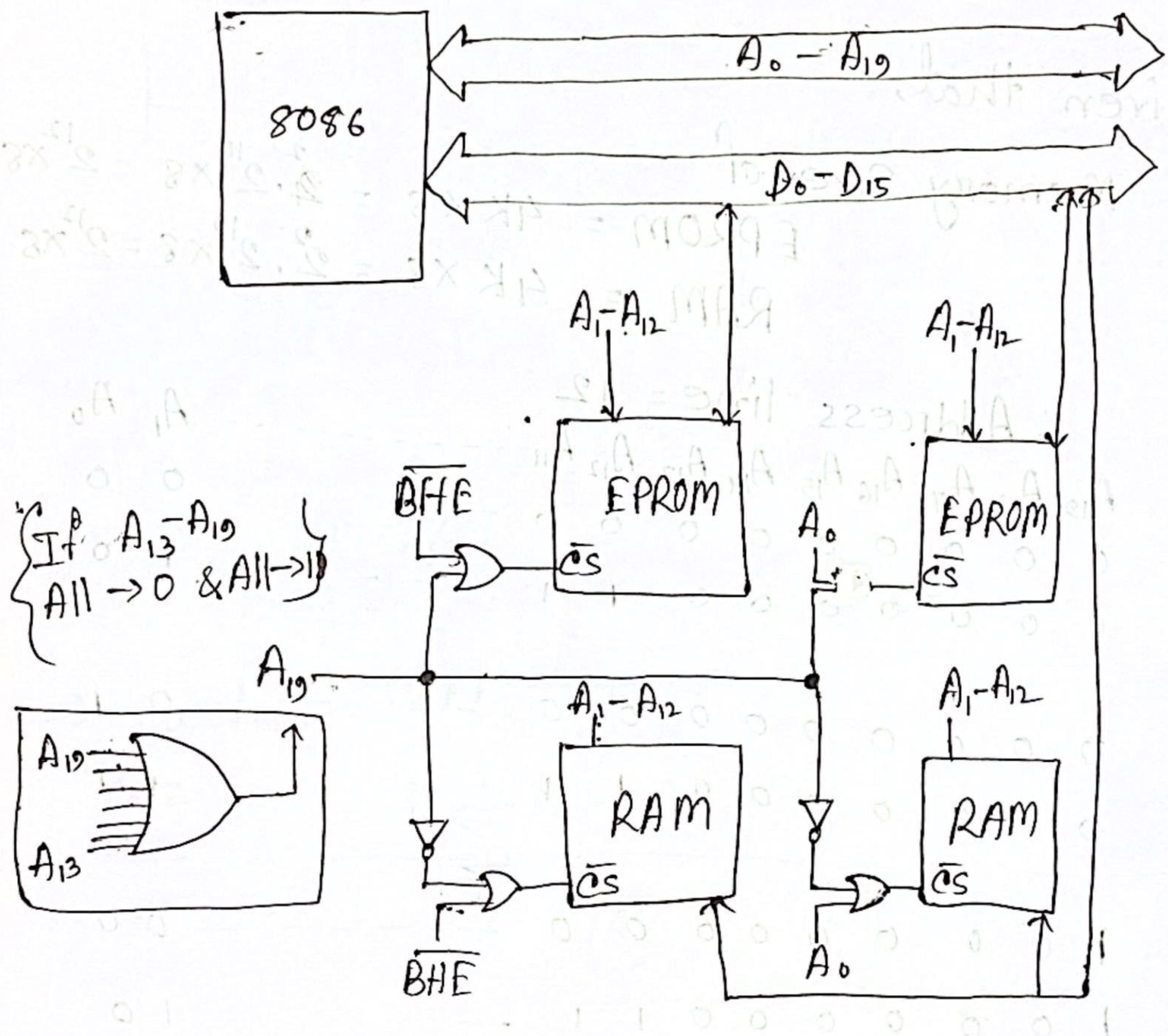
$$EPROM = 4K \times 8 = 2^2 \cdot 2^{10} \times 8 = 2^{12} \times 8$$

$$RAM = 4K \times 8 = 2^2 \cdot 2^{10} \times 8 = 2^{12} \times 8$$

$\therefore$ Address line = 12

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	----	A_1	A_0
EPROM-I	0	0	0	0	0	0	0	0	0	----	0	0
	0	0	0	0	0	0	0	1	1	----	1	0
EPROM-II	0	0	0	0	0	0	0	0	0	----	0	1
	0	0	0	0	0	0	0	1	1		1	1
RAM-I	1	0	0	0	0	0	0	0	0	----	0	0
	1	0	0	0	0	0	0	1	1	----	1	0
RAM-II	1	0	0	0	0	0	0	0	0	----	0	1
	1	0	0	0	0	0	0	1	1	----	1	1

Diagram;



8086
$A_0 - A_{19}$
$D_0 - D_{15}$
$A_1 - A_{12}$
EPROM
$\overline{CS}$
$\overline{BHE}$
A_0
If $A_{13} - A_{19}$
All → 0 & All → 1
A_{19}
A_{13}
RAM

P-2: Interface 32KB of RAM memory to the 8086 microprocessor system using absolate decoding with the suitable address.

Solⁿ:

Given that,

Total RAM = 32 KB

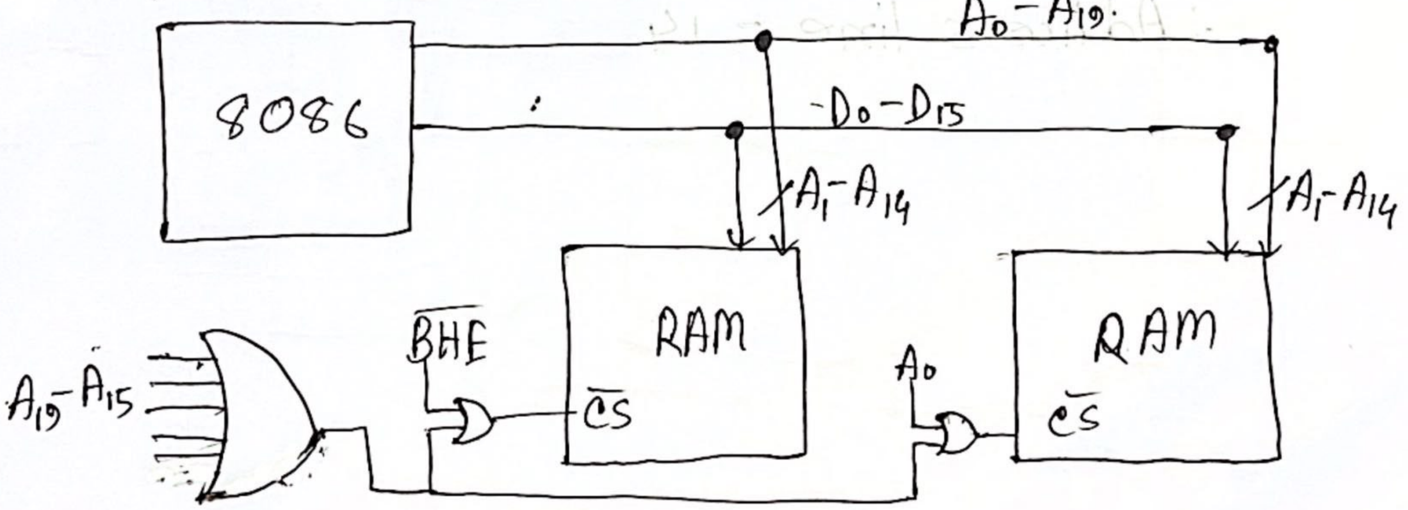
$\therefore$ Even and odd Bank $= \frac{32}{2}$ KB $= 16$ KB

$= 2^4 . 2^{10} \times 8 = 2^{14} \times 8$

$\therefore$ Address Line = 14

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	----	A_1	A_0
									0	0
RAM-I	0	0	0	0	0	0	0		1	0
	0	0	0	0	0	1	1			
RAM-II	0	0	0	0	0	0	0	----	0	1
	0	0	0	0	0	1	1		1	1

Diagram



P-3: Interface 32K word of memory to the 8086 microprocessor system. Available memory chips are $16K \times 8$ RAM use suitable decoder for generating chip select logic.

Sol^n:

we know that,

$$\text{word} = 16 \text{ bits} = 2B$$

Required memory

$$32K \text{ words} = 32K \times 2 \; B$$
$$= 64KB$$

$\because$ Eeach RAM chip $= 16K \times 8 = 16KB$

$\therefore$ so, nomber of chips required $= \dfrac{64KB}{16KB}$

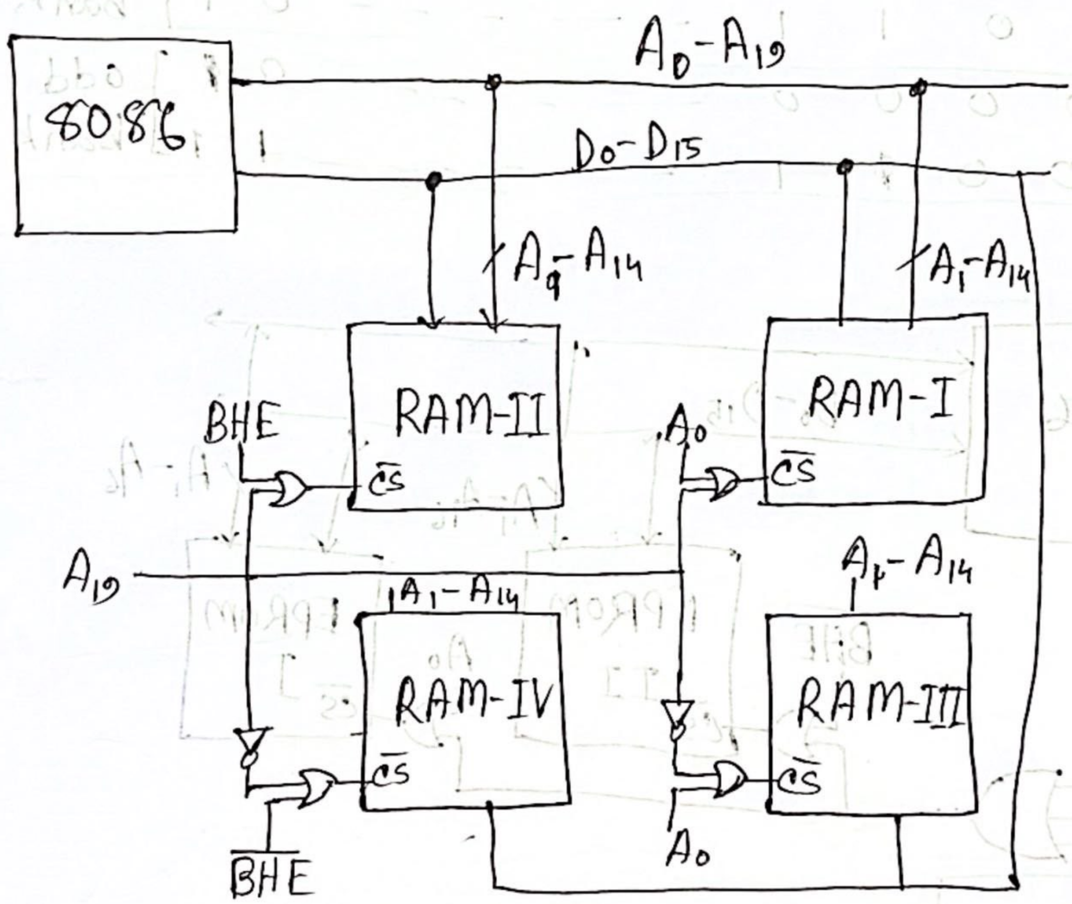
$$= 4 \text{ chips}$$

$$RAM \rightarrow 16KB = 2^4 \cdot 2^{10} \times 8 = 2^{14} \times 8$$

$\therefore$ Address line $= 14$

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	---	A_1	A_0
RAM-I	0	0	0	0	0	0	0	---	0	0
	0	0	0	0	0	1	1	---	1	0
RAM-II	0	0	0	0	0	0	0	---	0	1
	0	0	0	0	0	1	1	---	1	1
RAM-III	1	0	0	0	0	0	0	---	0	0
	1	0	0	0	0	1	1	---	1	1
RAM-IV	1	0	0	0	0	0	0	---	0	1
	1	0	0	0	0	1	1	---	1	1

Diagram:



P-4: Interface microprocessor 8086 with two number of IC-27512 chip.

Sol^n:

Given that,

2 number of IC-27512 → EPROM

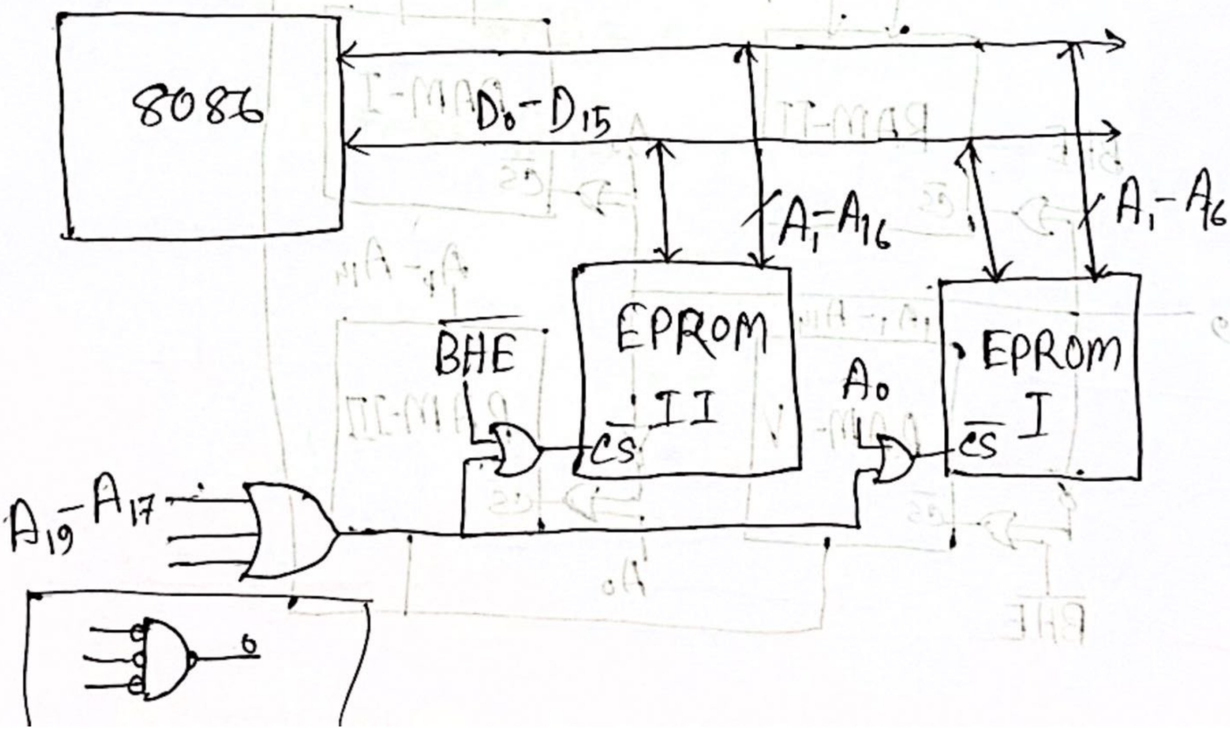
$\therefore$ memory size $= \frac{512}{8}$ KB $= 64$ KB

$= 2^6 \times 8$

$\therefore$ Address line $= 16$ $\qquad = 2^{16} \times 8$

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	---	A_1	A_0	
EPROM -I	0	0	0	0	0	---	0	0	Even bank
	0	0	0	1	1	---	0	1	
EPROM -II	0	0	0	0	0	---	0	1	odd bank
	0	0	0	1	1	---	1	1	

Diagram



P-5: Interface 16K×8 memory location for microprocessor 8086. The starting of the microprocessor is C0000 H.

Soln: Given that, starting address = C0000 H

Total memory location 16K×8

For even and odd bank $= \frac{16\,KB}{2} = 8KB$

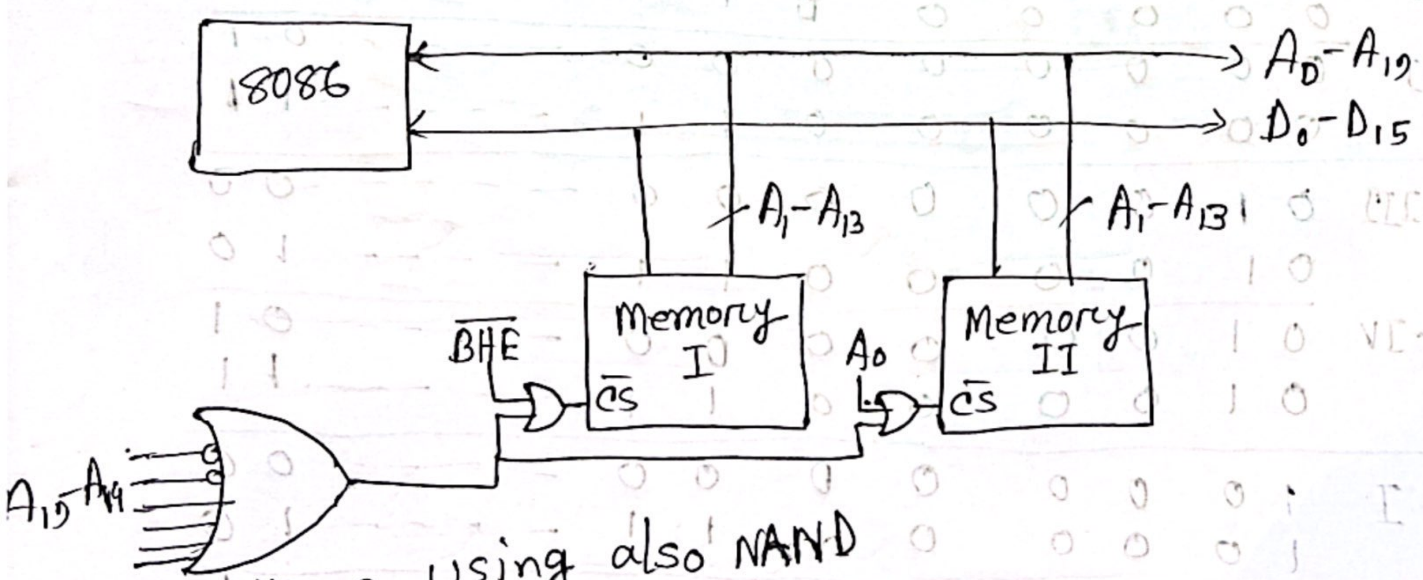
$= 2^3 . 2^{10} \times 8$

$= 2^{13} \times 8$

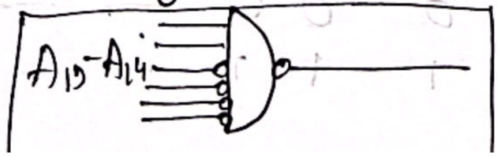
Address line = 13

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	------	A_1	A_0	
Memory I	1	1	0	0	0	0	0	0	------	0	0	Even bank
	1	1	0	0	0	0	1	1	------	1	0	
Memory II	1	1	0	0	0	0	0	0	------	0	1	odd bank
	1	1	0	0	0	0	1	1	------	1	1	

Diagram



Here using also NAND



P-61 Interface 32KB ROM using 8KB ROM and 32 KB RAM using 16 KB chips with 8086.

Soln:

Given that.

Total memory 32KB

For ROM:

Required no. of ROM $= \frac{32\,KB}{8\,KB} = 4$

memory size $= 8KB = 2^3 \cdot 2^{10} \times 8 = 2^{13} \times 8$

$AL = 13$

For RAM:

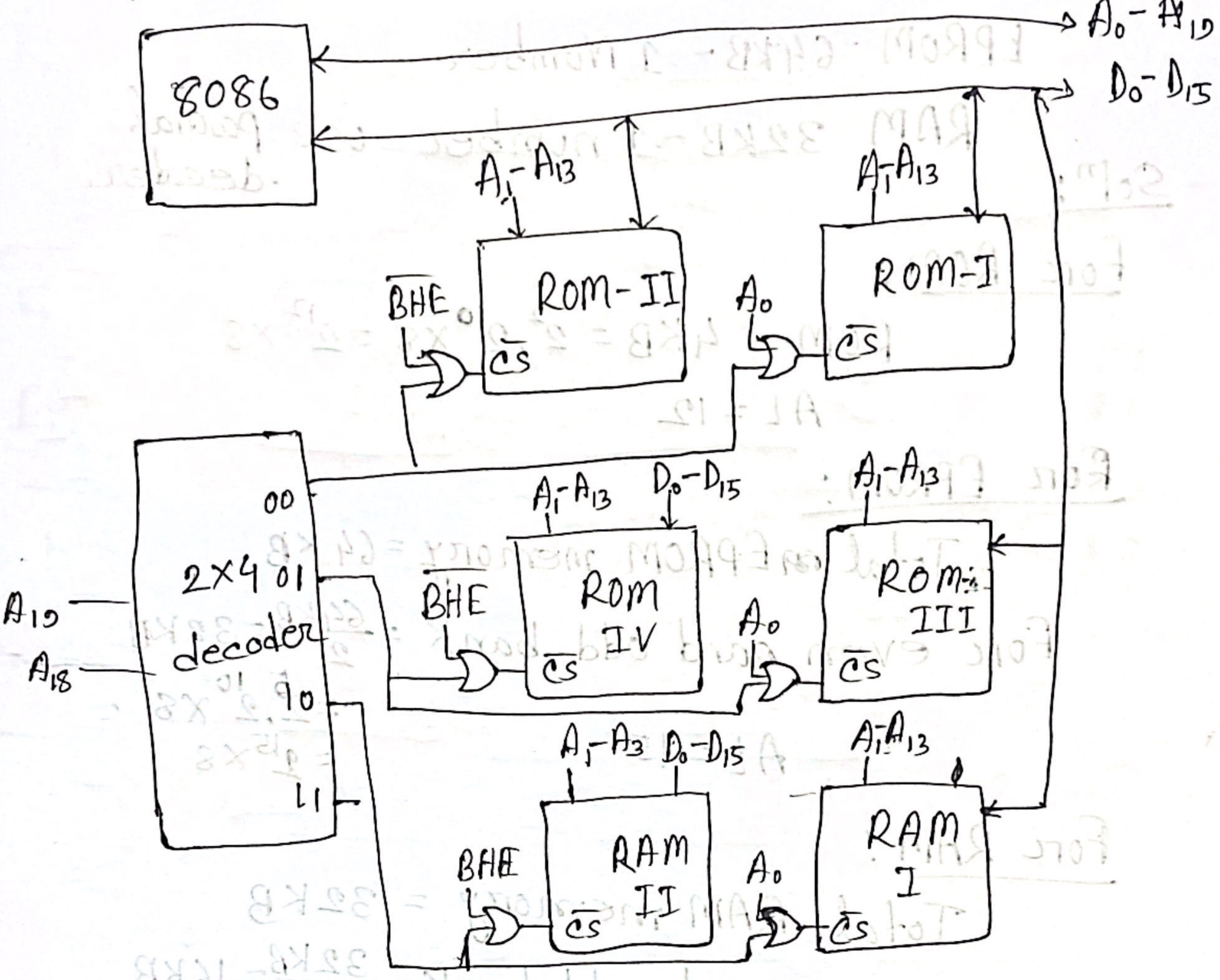
Required no. of RAM $= \frac{32KB}{16KB} = 2$

Memory size $= 16\,KB = 2^4 \cdot 2^{10} \times 8 = 2^{14} \times 8$

$AL = 14$

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	----	A_1	A_0
ROM-I	0	0	0	0	0	0	0	0	----	0	0
	0	0	0	0	0	0	1	1	----	1	0
ROM-II	0	0	0	0	0	0	0	0	----	0	1
	0	0	0	0	0	0	1	1	----	1	1
ROM-III	0	1	0	0	0	0	0	0	----	0	0
	0	1	0	0	0	0	1	1	----	1	0
ROM-IV	0	1	0	0	0	0	0	0	----	0	1
	0	1	0	0	0	0	1	1	----	1	1
RAM-I	1	0	0	0	0	0	0	0	----	0	0
	1	0	0	0	0	1	1	1	----	1	0
RAM-II	1	0	0	0	0	0	0	0	----	0	1
	1	0	0	0	0	1	1	1	----	1	1

Diagram:


$A_0 - A_{19}$
$D_0 - D_{15}$
8086
$A_1 - A_{13}$
ROM-II
$\overline{BHE}$
$\overline{CS}$
A_0
ROM-I
2X4 decoder
00
01
10
11
A_{19}
A_{18}
ROM IV
ROM III
BHE
$D_0 - D_{15}$
RAM II
RAM I
$A_1 - A_{13}$

P-71 ROM 4KB 2 Number

EPROM 64KB - 1 Number

RAM 32KB - 1 number use partial decoder.

Soln:

For ROM:

$ROM \rightarrow 4KB = 2^2 . 2^{10} \times 8 = 2^{12} \times 8$

$AL = 12$

For EPROM:

Total EPROM memory = 64 KB

For even and odd bank $= \frac{64KB}{2} = 32KB$

$= 2^5 . 2^{10} \times 8$

$= 2^{15} \times 8$

$AL = 15$

For RAM:

Total RAM memory = 32KB

For even and odd bank $= \frac{32KB}{2} = 16KB$

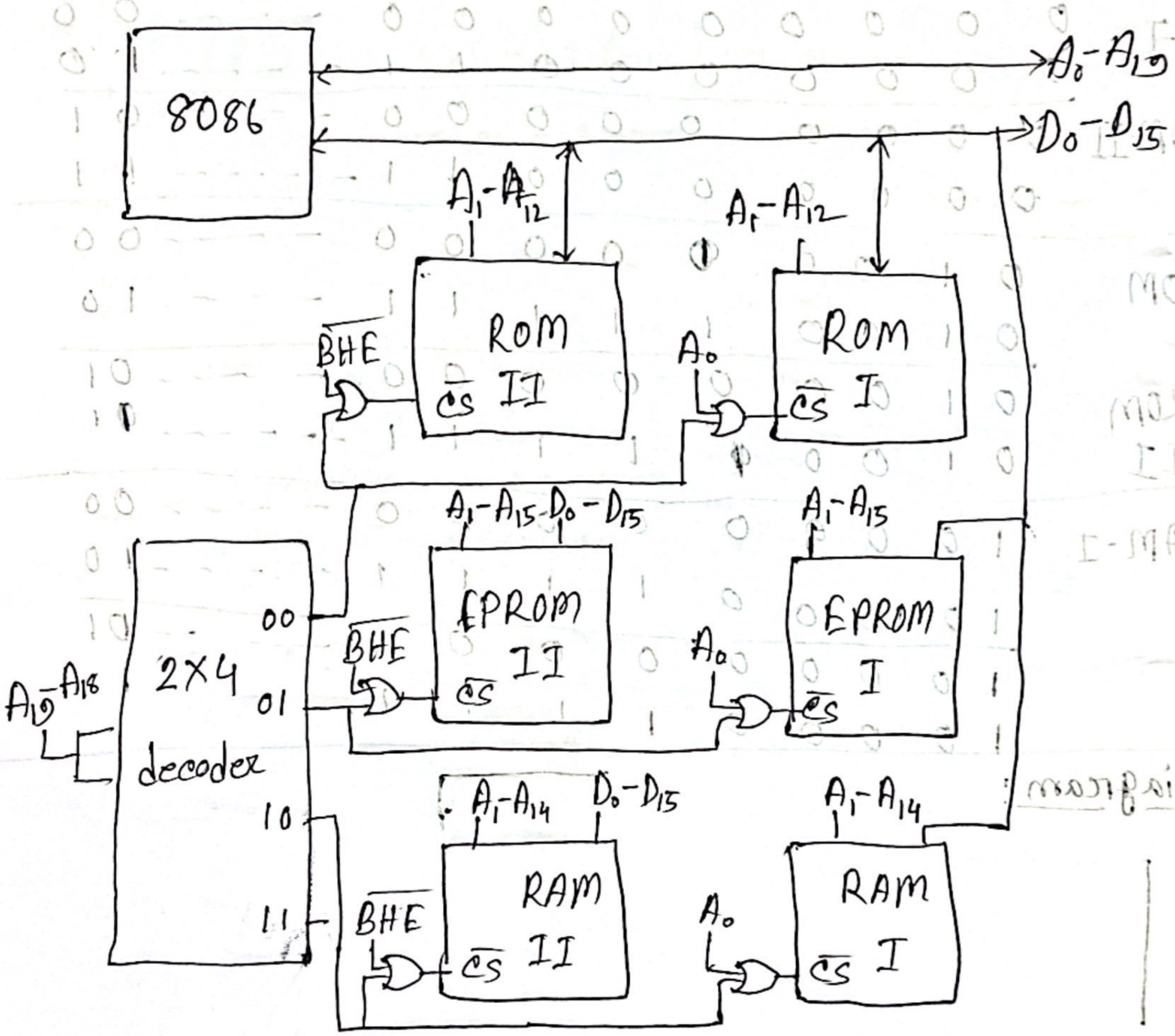
$= 2^4 . 2^{10} \times 8$

$= 2^{14} \times 8$

$AL = 14$

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	-----	A_1	A_0
ROM-I	0	0	0	0	0	0	0	0	0	-----	0	0
	0	0	0	0	0	0	0	1	1	-----	1	0
ROM-II	0	0	0	0	0	0	0	0	0	-----	0	1
	0	0	0	0	0	0	0	1	1	-----	1	1
EPROM I	0	1	0	0	0	0	0	0	0	-----	0	0
	0	1	0	0	1	1	1	1	1	-----	1	0
EPROM II	0	1	0	0	0	0	0	0	0	-----	0	1
	0	1	0	0	1	1	1	1	1	-----	1	1
RAM-1	1	0	0	0	0	0	0	0	0	-----	0	0
	1	0	0	0	0	1	1	1	1	-----	1	0
	1	0	0	0	0	0	0	0	0	-----	0	1
	1	0	0	0	0	1	1	1	1	-----	1	1

Diagram:



8086
A0-A19
D0-D15
A1-A12
A1-A12
ROM II
ROM I
BHE
A0
CS
A1-A15
D0-D15
A1-A15
EPROM II
EPROM I
2X4 decoder
A19-A18
00
01
10
11
A1-A14
D0-D15
A1-A14
RAM II
RAM I